Marc Hermes

Curriculum Vitae

hermes.marc@gmail.com
hermesmarc.github.io

Education

Oct. 2022 – Mar. 2024 **PhD candidate in Computer Science**, Radboud University, Nijmegen, Netherlands

2018–2022 **Master of Science in Mathematics**, *Saarland University*, Saarbrücken, Germany, GPA: 1.3 (Germany)

- Achieved top grade (1.0) MA Thesis, resulting in an award-winning conference paper and journal publication.
- Relevant Coursework: Machine Learning, Image Processing and Computer Vision, Probability Theory, Stochastic Processes, Random Matrices.

2013–2017 **Bachelor of Science in Physics**, *Saarland University*, Saarbrücken, Germany, GPA: 2.2 (Germany)

- Analyzing images (ImageJ, Python) and other experimental data (Python), modelling physical systems (C, MATLAB).

2010–2013 **Primary Education**, *Lycée Classique d'Echternach*, Echternach, Luxembourg, GPA: Good (Luxembourg)

2006–2010 **Secondary Education**, *Lycée Technique Joseph Bech*, Grevenmacher, Luxembourg

Work Experience

Dec. 2024 – May 2025 **Junior Data Scientist**, *Machine Learning Programs*, Amsterdam, Netherlands

- Experimenting on how the inclusion of weather data impacts model prediction for claim rates on private car policies.
- Conducting feature selection to identify the most impactful features for a new claim prediction model.
- Designing and creating tests for evaluating the performance of a LLM on a classification task.
- Starting development on a client facing API using FastAPI.
- Reviewing code and validating merge requests.

2022–2024 **PhD candidate (Computer Science)**, *Radboud University*, Nijmegen, Netherlands

- Conducted and published research on the automated program verification for data structures related to the C programming language.
- Helping students with their course material in practical sessions, grading exams and supervising students for giving seminar talks.

2022–2024 **Voluntary Service**, *PhD Organization Nijmegen*, Nijmegen, Netherlands

- Organizing and hosting events for PhD students as a member of the social committee.

2017–2022 **Student Teaching Assistant**, *Saarland University*, Saarbrücken, Germany

- Leading tutorial and exercise sessions for numerous courses in Mathematics, grading exams and lecturing.

2017–2018 **Voluntary service**, *Saarland University*, Chairman of the Physics student council.

Skills and Strengths

- **Programming:** Python (Pandas, Sklearn, Seaborn, FastAPI), Git, Emacs, SQL, C, OCaml, SML, Racket, Rocq, LaTeX, Shell scripting.
- Familiarity with both object-oriented and functional programming paradigms.
- **Natural Languages:** German (native), Luxemburgish (native), English (proficient), French (fluent) and Dutch (beginner).
- Successfully transitioning between related fields (Physics, Mathematics, Computer Science, Data Science) and adapting new domain-specific ways of thinking.
- Effective and clear cross-disciplinary communication to experts and non-experts.
- Very strong analytical and logical skills.

Awards

2022 **Best Paper by Junior Researcher**, Issued by the FSCD Committee for our paper *"An Analysis of Tennenbaum's Theorem in Constructive Type Theory"*

Scientific Research

2024 **Conference Publication**, *Modular Verification of Intrusive List and Tree Data Structures in Separation Logic*, ITP 2024, Marc Hermes, Robbert Krebbers

2024 **Journal Publication**, *An Analysis of Tennenbaum's Theorem in Constructive Type Theory*, Logical Methods in Computer Science, Dominik Kirst, Marc Hermes

2023 **Journal Publication**, *Synthetic Undecidability and Incompleteness of First-Order Axiom Systems in Coq (Extended Version)*, Journal of Automated Reasoning, Dominik Kirst, Marc Hermes

2022 **Conference Publication**, *An Analysis of Tennenbaum's Theorem in Constructive Type Theory*, FSCD 2022, Marc Hermes, Dominik Kirst

2021 **Master Thesis**, *Modeling Peano Arithmetic in Constructive Type Theory*, Saarland University, Department of Mathematics, under supervision of Prof. Dr. Moritz Weber. (Grade: 1.0)

2021 **Conference Publication**, *Synthetic Undecidability and Incompleteness of First-Order Axiom Systems in Coq*, ITP 2021, Dominik Kirst, Marc Hermes

2017 **Bachelor Thesis**, *Quantum memory for photons*, Saarland University, Physics Department, under supervision of Prof. Dr. Giovanna Morigi. (Grade : 2.0)